

Dual-Mode AI System for Accurate Plant Disease Detection and Smart Farm Recommendations

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Project Guide



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Overview

- Introduction
- Problem Statement
- Literature Review
- System Architecture
- Description of Dataset
- Implementation
- Remaining Work with timeline
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Introduction

Cloud-Based AI System for Real-Time Crop Disease Detection and Farm Advisory with offline Mode

- **Traditional Diagnosis is Inefficient** - Methods relying on manual inspection or sending samples to labs are slow, require significant expertise, and delay crucial intervention, often leading to widespread yield loss
- **Leveraging AI and Cloud**-By integrating deep learning models with scalable cloud infrastructure, mobile applications can democratize expert diagnostic services, providing instant analysis and actionable advice directly to the farmer.
- **The Need for Speed and Scale**-Early and accurate diagnosis is essential. We need a system that can provide expert-level identification and management advice immediately at the farm site, regardless of the user's technical knowledge.



Problem Statement

To build a mobile app for crop disease diagnosis that delivers fast, expert results using cloud-based AI when online, and provides instant, basic detection with an on-device model when offline—ensuring farmers get reliable help anywhere, anytime.



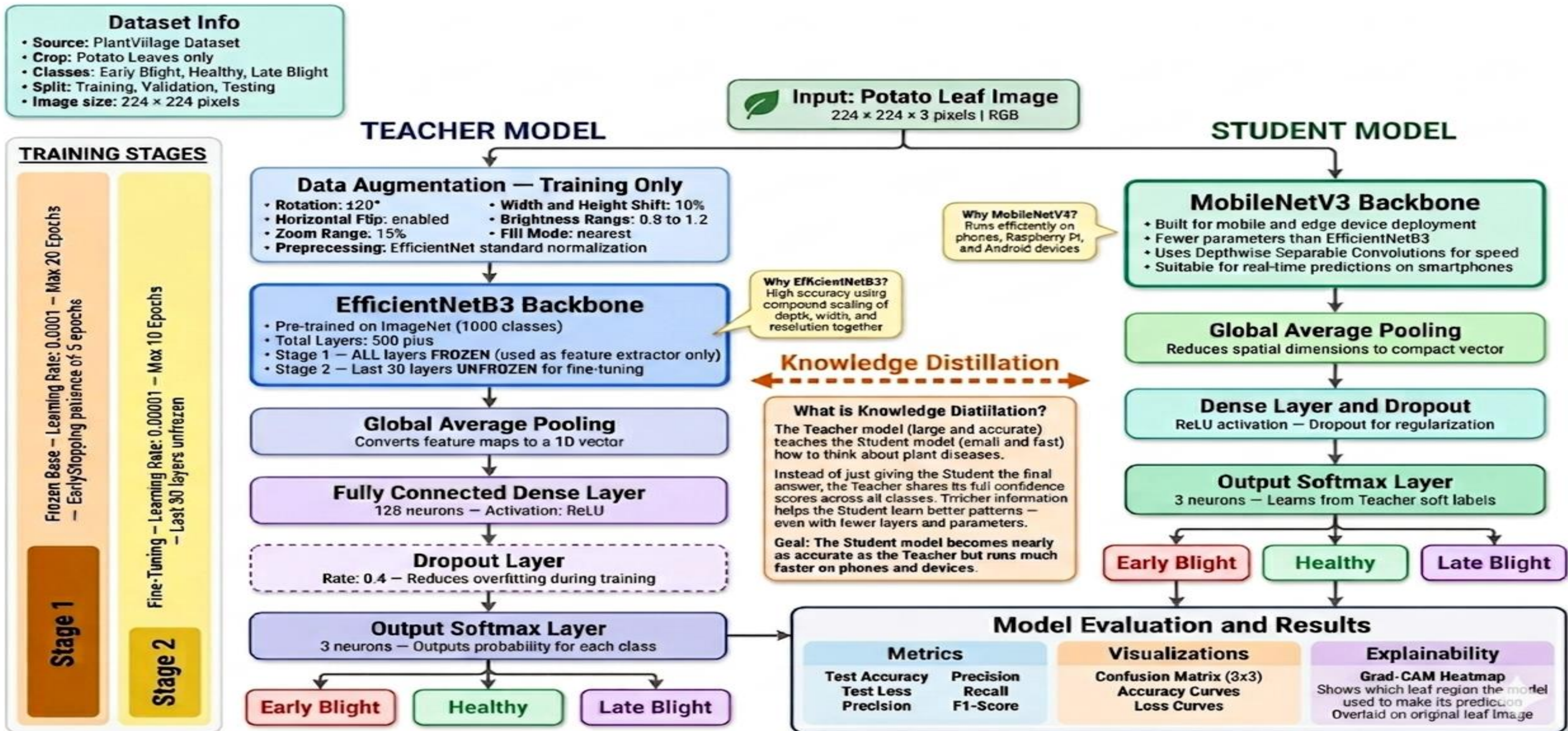
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Literature Review

Category	Cloud-Based Systems	Lightweight (Lite) Models	Hybrid Systems (Online + Offline)
Reference Papers	[1],[2],[5]	[3],[4],[6]	None Available
Deployment	Cloud/Web server	Mobile & Edge devices	Cloud + Mobile
Key Features	Cloud inference, Weather API, Disease forecasting, Farmer advisory, Marketplace integration.	Depthwise Separable Conv, SE Attention, Residual Blocks, Knowledge Distillation, Grad-CAM	Automatic online/offline switching, Model synchronization, Offline logging
Advantages	High accuracy, scalable computing, easy model updates, rich agricultural services	Fast inference, low memory, offline operation, suitable for smartphones	Combines cloud intelligence with offline accessibility
Limitations	Requires continuous internet, cloud hosting cost, unusable in poor connectivity	Limited contextual information , no weather/outbreak monitoring , slight performance loss after compression	No existing implementation found in surveyed literature
Research Gap	Cannot work offline	Cannot utilize cloud services	Opportunity to develop a dual-mode hybrid system

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System Architecture



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Description of Dataset

	Name	Images	Size of Image	Type	Dimension	Class
Dataset 1 (Potato)	<u>Potato Leaf Dataset(PLD)</u> <u>By Rizwan Saeed</u>	4072	7.04kb-11.05kb	jpeg	256X256 Pixels	Early_Blight Healthy Late_Blight
Dataset2 (Tomato)	<u>Plant Leaf Diseases Training Dataset</u>	4500	16.15kb-20.72kb	jpeg	256X256 Pixels	Early_Blight Healthy Late_Blight
Dataset3 (Rice)	<u>Plant Leaf Diseases Training Dataset</u>	3457	11.56KB-16.34Kb	Jpeg	256X256 Pixels	Blast Blight Brown spot Healthy

Implementation

Input	Models	Accuracy	Precision	Recall	F1-Score
Dataset 1 (Potato Leaf) _[7]	EfficientNetB3	0.9880	0.9851	0.9882	0.9866
	ResNet50	0.9832	0.9818	0.9843	0.9829
	DenseNet121	0.9784	0.9771	0.9797	0.9771
	MobileNetV3-Large	0.9827	0.9874	0.9679	0.9824
	EfficientNetB2	0.9704	0.9776	0.9679	0.9872
	ResNet152V2	0.9877	0.9901	0.9852	0.9877
	DenseNet201	0.9802	0.9815	0.9795	0.9803
	MobileNetV2	0.9866	0.9870	0.9860	0.9866



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Implementation

Input	Models	Accuracy	Precision	Recall	F1-Score
Dataset 2 (Tomato Leaf) ^[8]	EfficientNetB3	0.9483	0.95	0.947	0.948
	ResNet152V2	0.9867	0.989	0.985	0.987
	DenseNet201	0.9867	0.987	0.983	0.985
	MobilNetV3Large	0.9970	0.993	0.991	0.9970
	EfficientNetB2	0.9542	0.956	0.952	0.954
	ResNet50	0.8021	0.805	0.799	0.802
	MobileNetV2	0.9867	0.988	0.984	0.986
	DenseNet121	0.9792	0.9818	0.9761	0.9784



Implementation

Input	Models	Accuracy	Precision	Recall	F1-Score
Dataset 3 (Rice Leaf) ^[8]	EfficientNetB3	0.9806	0.9851	0.9882	0.9806
	ResNet50	0.8588	0.8903	0.8317	0.8929
	DenseNet201	0.9845	0.9771	0.9797	0.9839
	InceptionResNetV2	0.9451	0.9453	0.9357	0.9392
	EfficientNetB2	0.9806	0.9804	0.9791	0.9796
	ResNet152V2	0.9698	0.9725	0.9981	0.9702
	MobileNetv3Large	0.9354	0.9372	0.9311	0.9341
	MobileNetV2	0.9251	0.9278	0.9176	0.9226



Implementation

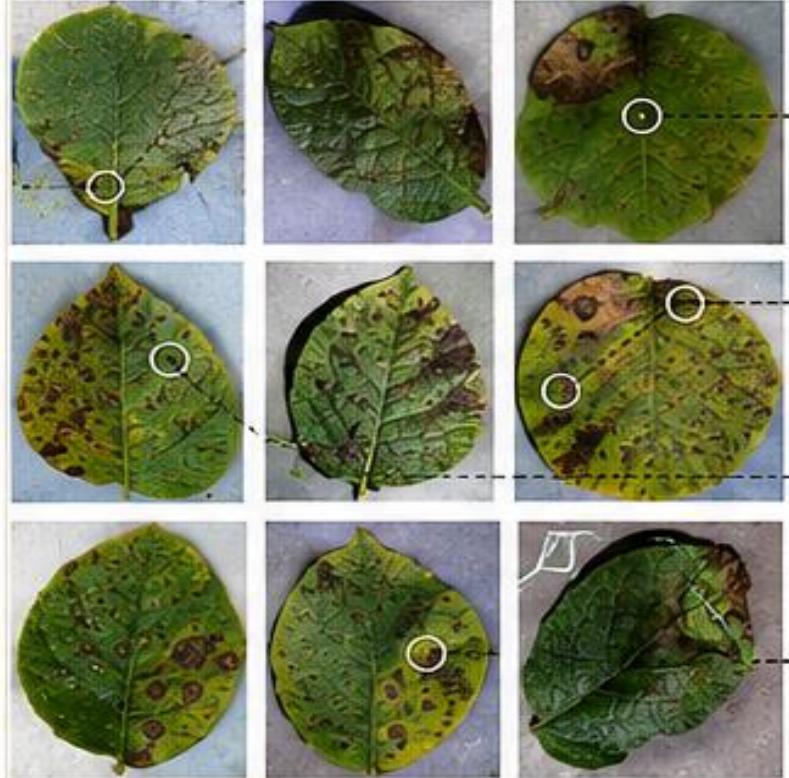
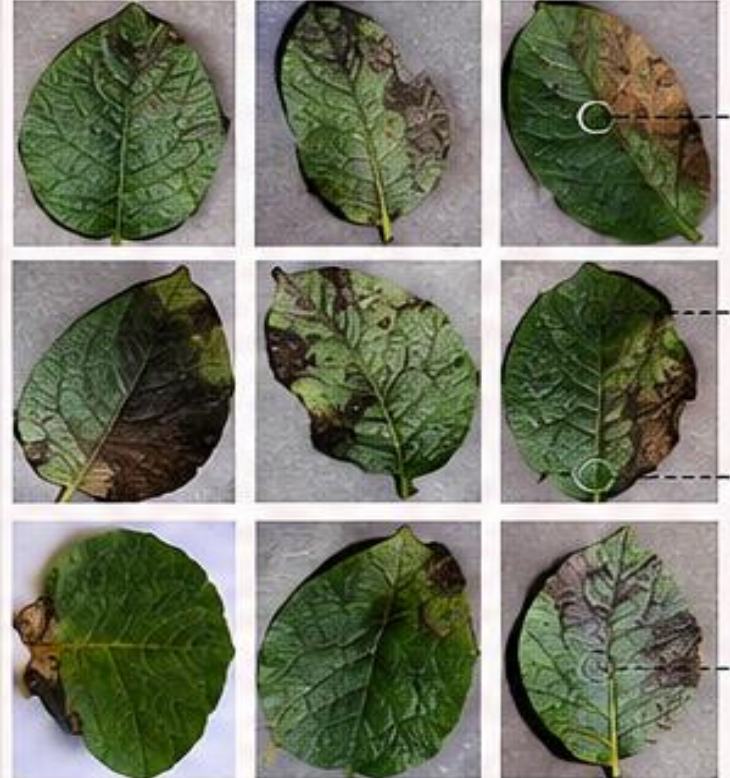
Conclusion:

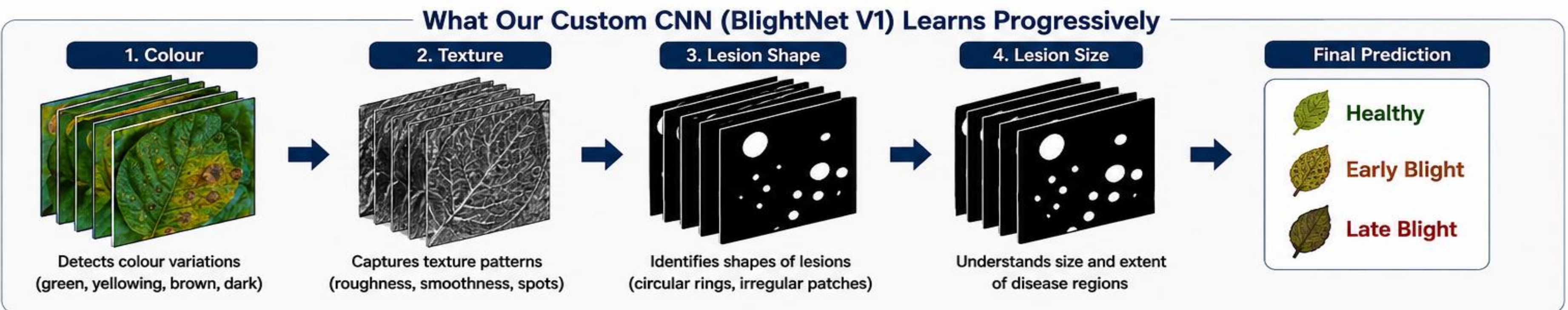
Among the evaluated models, EfficientNet-B3 achieved the highest overall classification performance, making it the most suitable choice for server-side (online) deployment, where computational resources are available and maximum accuracy is the priority. For mobile (offline) deployment, MobileNetV3 is the preferred model due to its lightweight architecture, fast inference speed, and low memory requirements while still maintaining good classification accuracy. To combine the strengths of both models, knowledge distillation can be applied, where EfficientNet-B3 acts as the teacher model and MobileNetV3 serves as the student model. This enables the mobile model to learn from the richer predictions of the larger network, improving its accuracy while remaining efficient enough for real-time smartphone deployment.



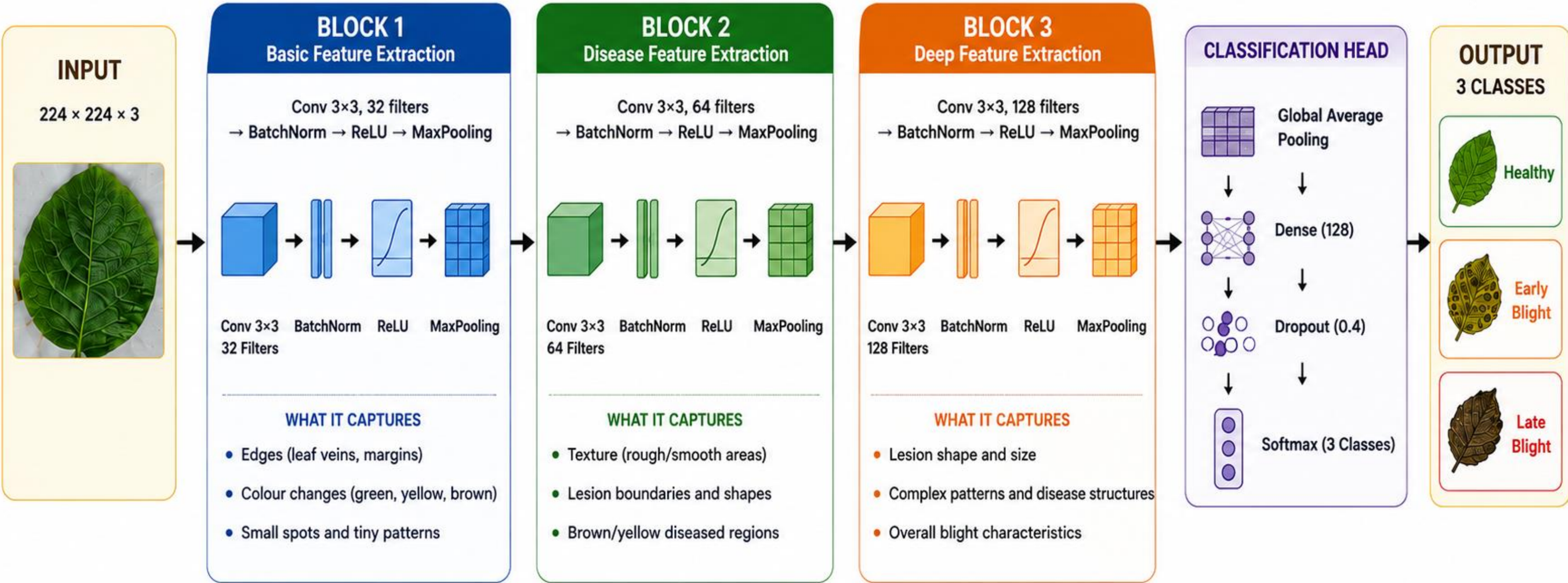
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Custom Model

Healthy	Early Blight	Late Blight
Representative Samples (from our dataset) 	Representative Samples (from our dataset) 	Representative Samples (from our dataset) 
Key Visual Characteristics ✓ Uniform green colour ✓ Smooth texture ✓ No lesions or spots ✓ Normal size and shape ✓ Even surface and colour distribution	Key Visual Characteristics - Brown circular lesions - Concentric rings (target like patterns) - Yellowing around lesions - Small to medium sized spots	Key Visual Characteristics - Large, irregular dark patches - Water-soaked appearance - Irregular lesion boundaries - Larger lesion size and rapid spread



BlightNet V1 – Low-Level Architecture (3 Convolution Blocks)

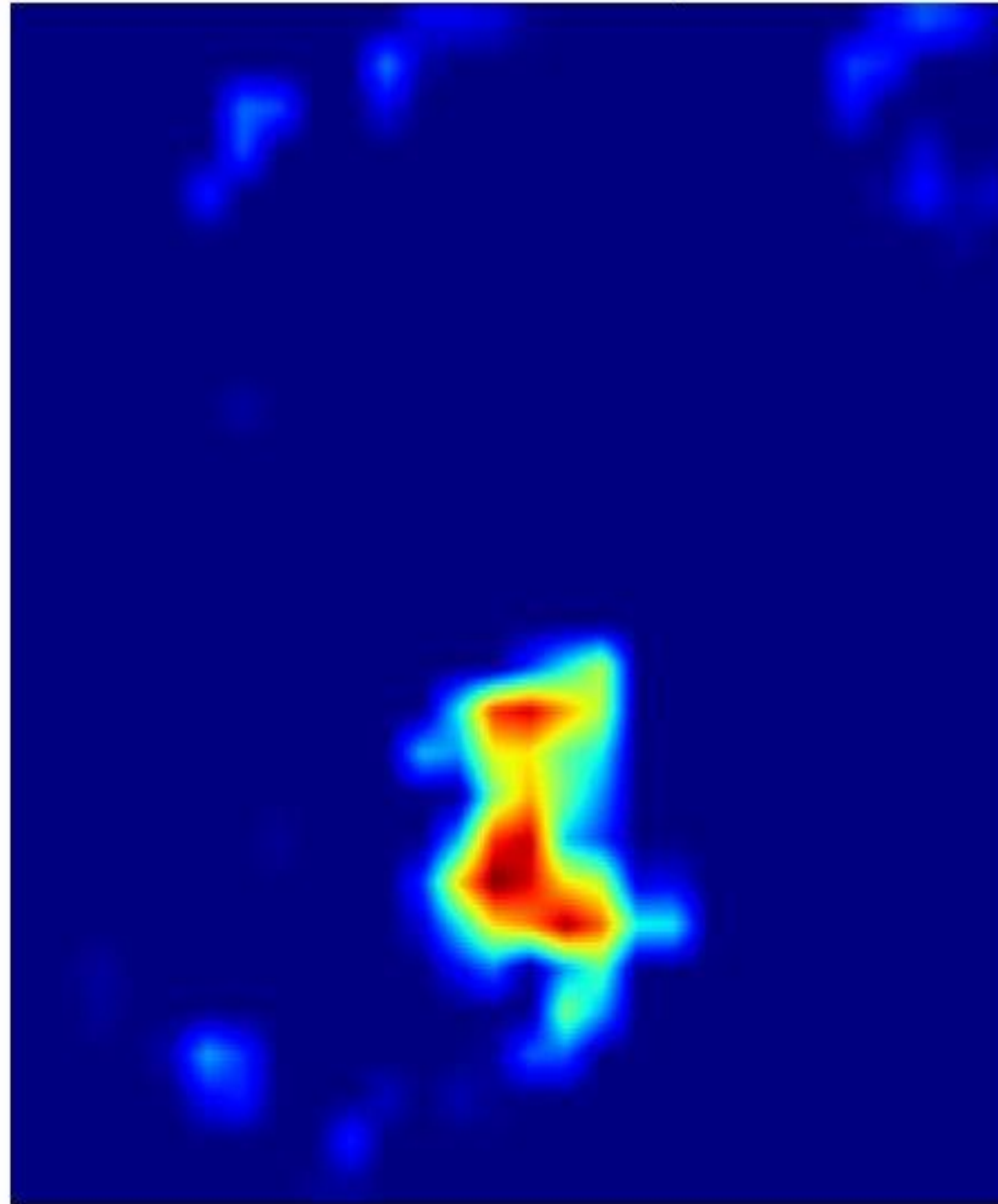


Implementation

Original Potato Leaf



Grad-CAM Heatmap



Grad-CAM Overlay
Late_Blight — 92.03%

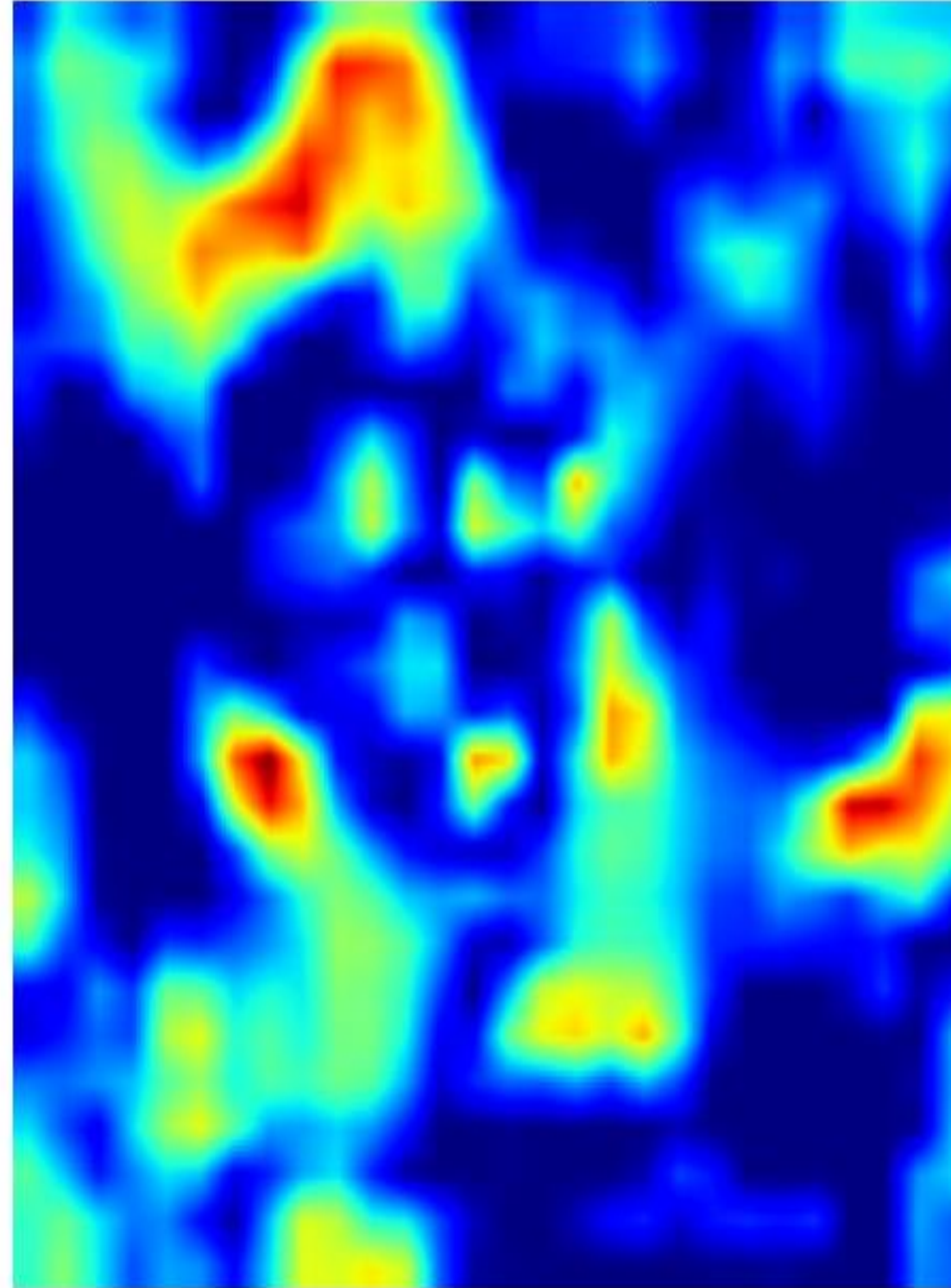


Implementation

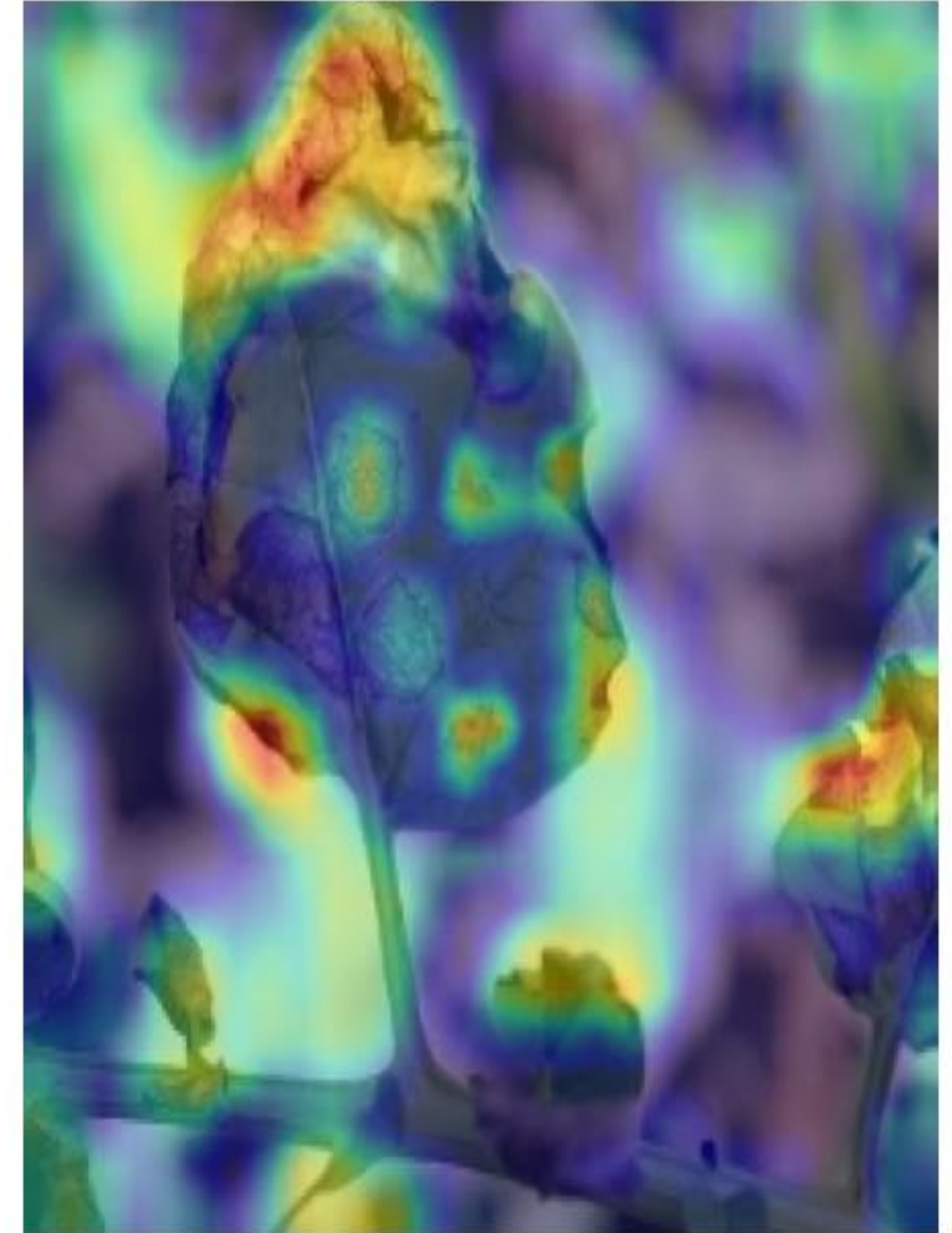
Original Potato Leaf



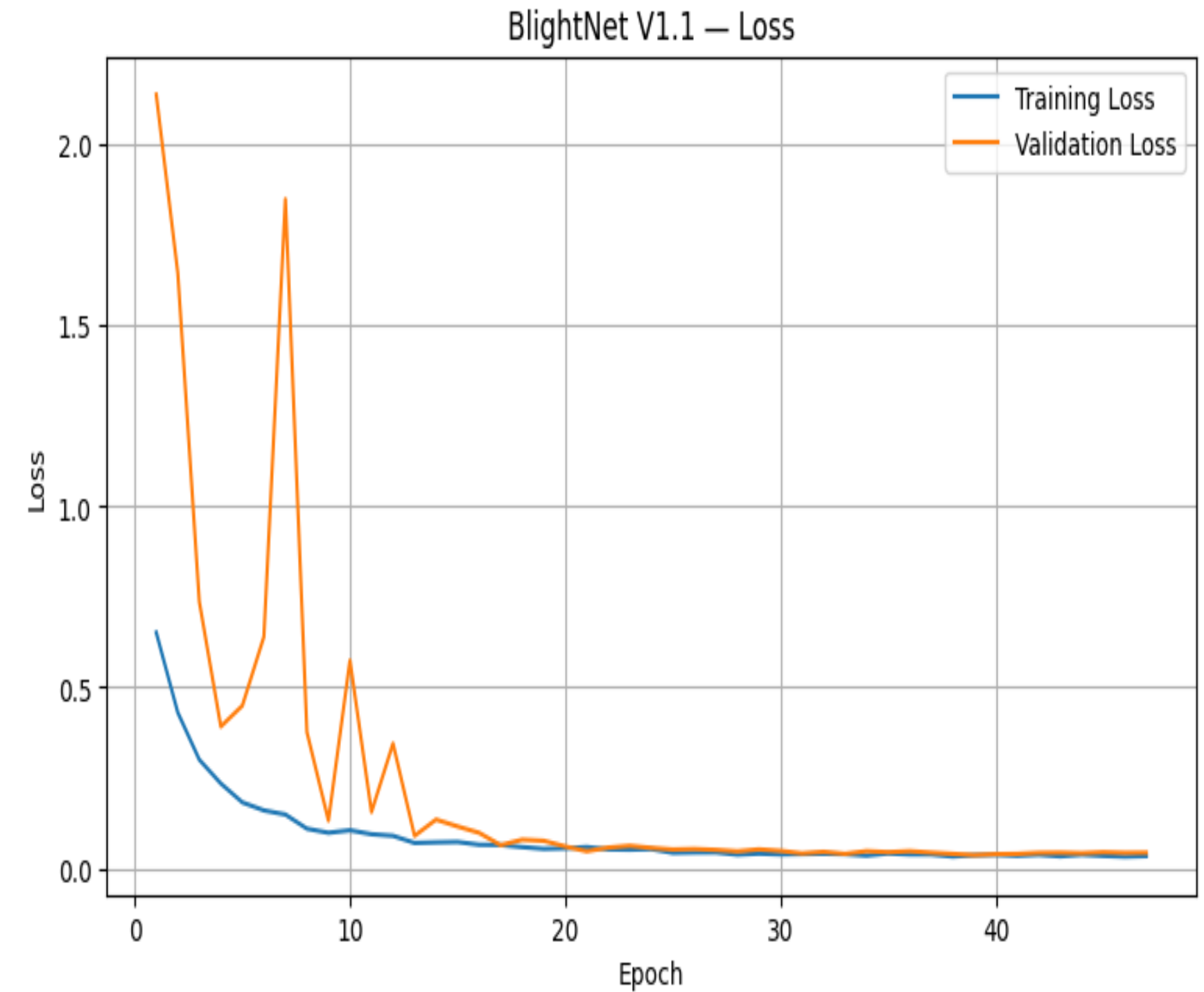
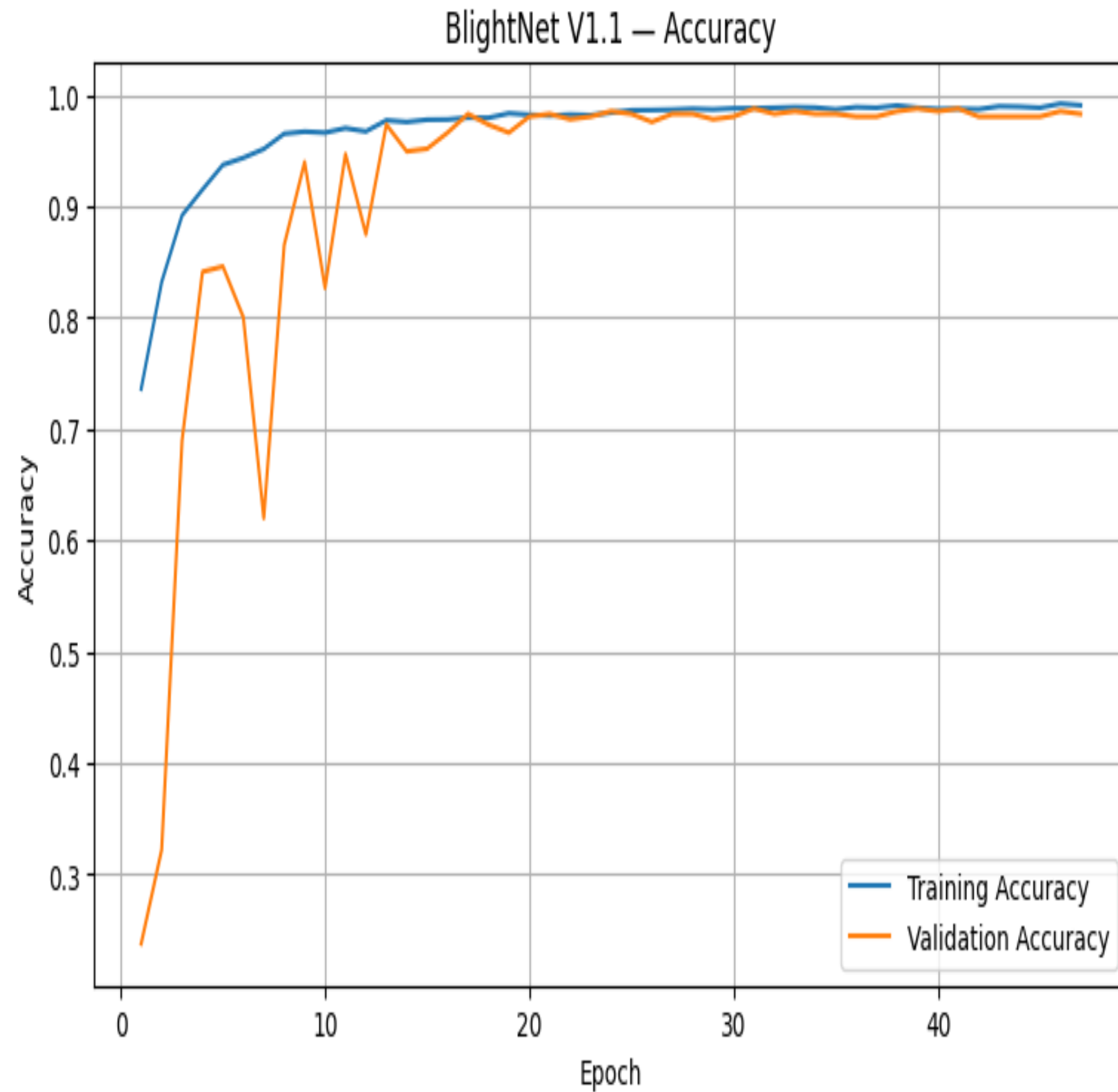
Grad-CAM Heatmap



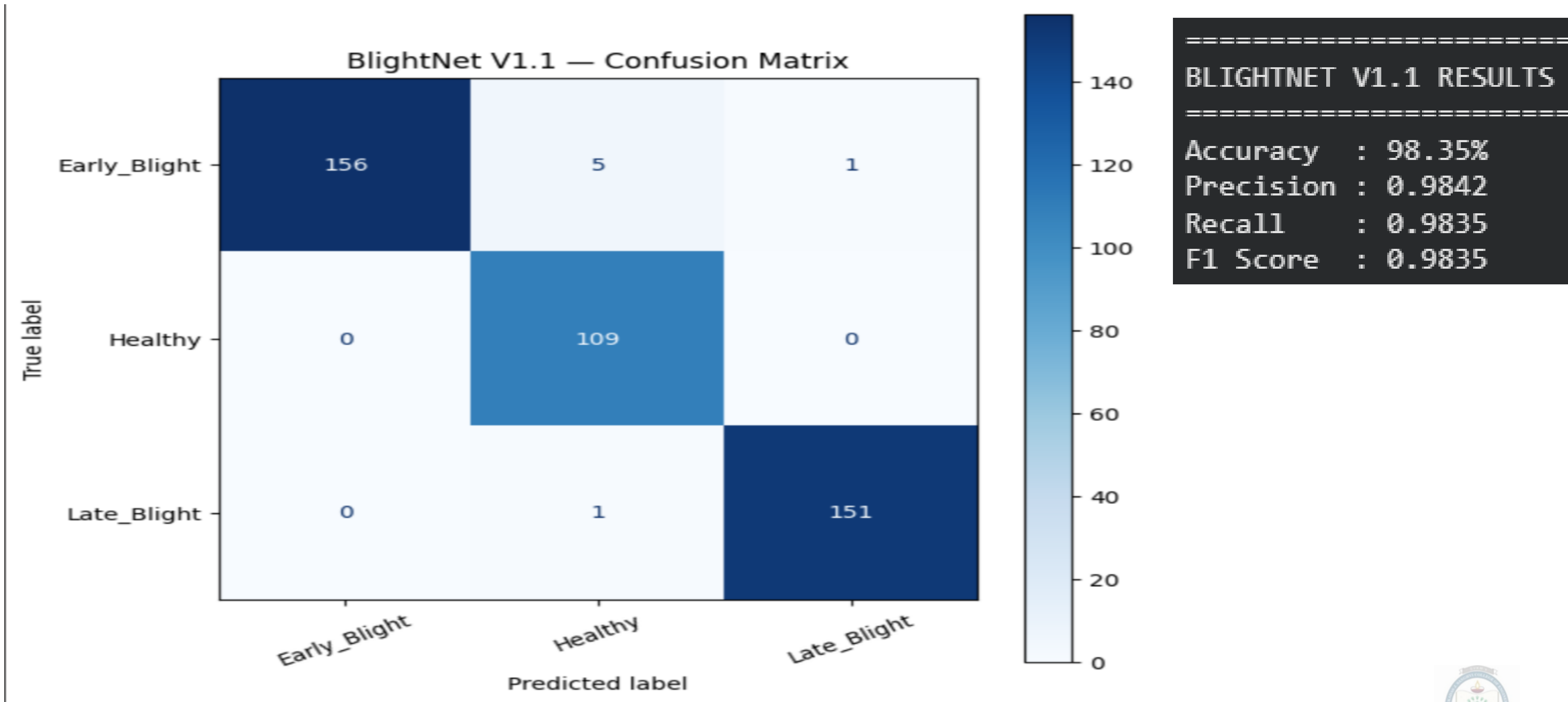
Grad-CAM Overlay
Early_Blight — 99.99%



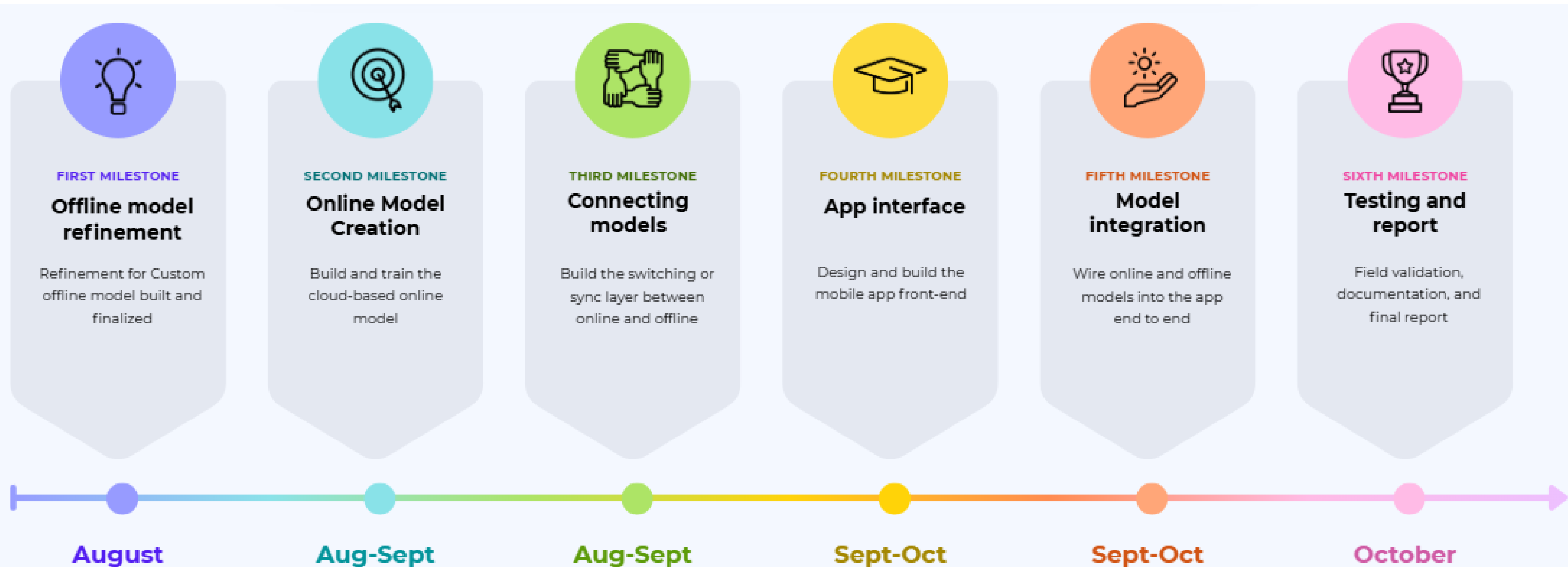
Implementation



Implementation



Remaining Work with Timeline



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Available at: <https://www.kaggle.com/datasets/nirmalsankalana/plant-diseases-training-dataset>



Thank You!!!

